

EXP. NO:3
MATLAB-BASED MODELLING OF VSB MODULATOR AND
DEMODULATOR FOR ANALOG TELEVISION
DATE: COMMUNICATION APPLICATIONS

Objectives:

- i) To generate and analyse the Vestigial Sideband (VSB) modulated signal by performing DSB-SC modulation and applying a suitable filter using MATLAB.
- ii) To implement coherent demodulation and recover the original message signal from the VSB signal, and compare the transmitted and recovered waveforms.

CODE

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;

% Message and carrier signals
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);

% DSB-SC modulation
dsb = m .* c;

% Approximate VSB using frequency-domain filtering
N = length(dsb);
f = (0:N-1)*(fs/N);
DSB_FFT = fft(dsb);

% Keep components around the carrier for demonstration
mask = (f >= fc-1000 & f <= fc+1000) | ...
       (f >= fs-(fc+1000) & f <= fs-(fc-1000));

VSB_FFT = DSB_FFT .* mask;
vsb = real(ifft(VSB_FFT));
```

```

% Coherent demodulation
demod = 2 * vsb .* c;

% Recover message using moving average
window = round(fs/1000);
rec = movmean(demod, window);

% Remove DC offset and normalize
rec = rec - mean(rec);
rec = rec / max(abs(rec));

% Plot results
figure('Name','DSB-SC and VSB','NumberTitle','off');

subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

subplot(5,1,3);
plot(t,dsb);
title('DSB-SC Signal');
grid on;

subplot(5,1,4);
plot(t,vsb);
title('Filtered DSB-SC Approximation');
grid on;
subplot(5,1,5);
plot(t,m,'b--');
hold on;
plot(t,rec,'r');
hold off;
title('Original and Recovered Message');
legend('Original','Recovered');
grid on;

```

OUTPUT

